

Hybrid H₂ Fuel Cell & Battery UAV

Dynamics, Power-Split & Flight Control

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Course: | Dynamic System Modelling (702MH0C023)

Category: | B (Aerial Dynamics) **Complexity:** | High

COs Exercised: | [CO1, CO2, CO3, CO4, CO5]

Date: | [19/04/2026]

System Introduction & Motivation

- **What is the system?**

A quadrotor UAV powered by a hybrid H_2 fuel cell + Li-ion battery bus. Four brushed-DC-motor-rotor pairs generate thrust; a cascaded flight controller stabilises it.

- **Why does it matter?**

H_2 UAVs promise -3x endurance over Li-ion drones but fuel cells respond slowly (seconds). Batteries are fast but heavy. A hybrid bus gets both: endurance from H_2 agility from the battery.

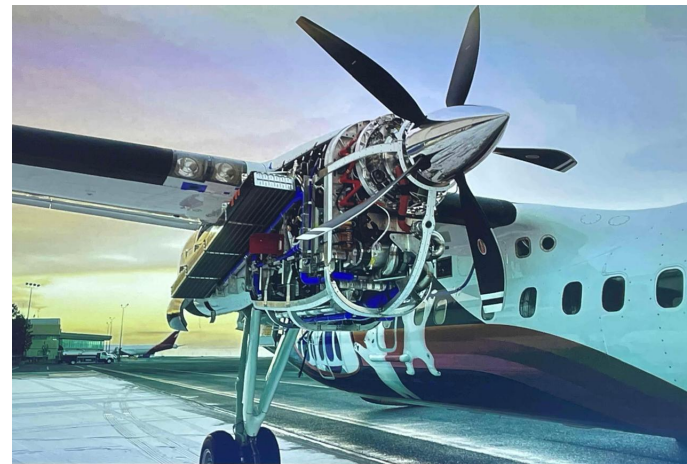
- **PBL Question :** Can we model 4 coupled physics domains – electrochemical, electrical, electromechanical, rigid-body-6-DoF – in one simulation, and show that the battery absorbs the fast transients while the fuel cell supplies the steady hover power?

System Introduction & Motivation

Applications :

Real-world inspiration: Universal Hydrogen's **De Havilland Canada Dash 8-300** hydrogen-electric demonstrator – a *megawatt-scale* fuel cell propulsion system flight-tested from Moses Lake, Washington (March 2023). It proves the exact same hybrid-bus concept we are modelling, but at full regional-aircraft scale.

- **Why this scales down to UAVs:** the underlying physics — slow fuel cell + fast battery on a common DC bus — is identical whether it drives a 25-ton turboprop or a 1.5 kg quadrotor. The UAV scale lets us model every coupled subsystem end-to-end in simulation and validate it on a benchtop.
- **Near-term deployment targets:** last-mile delivery, long-endurance surveillance, rural medical logistics.



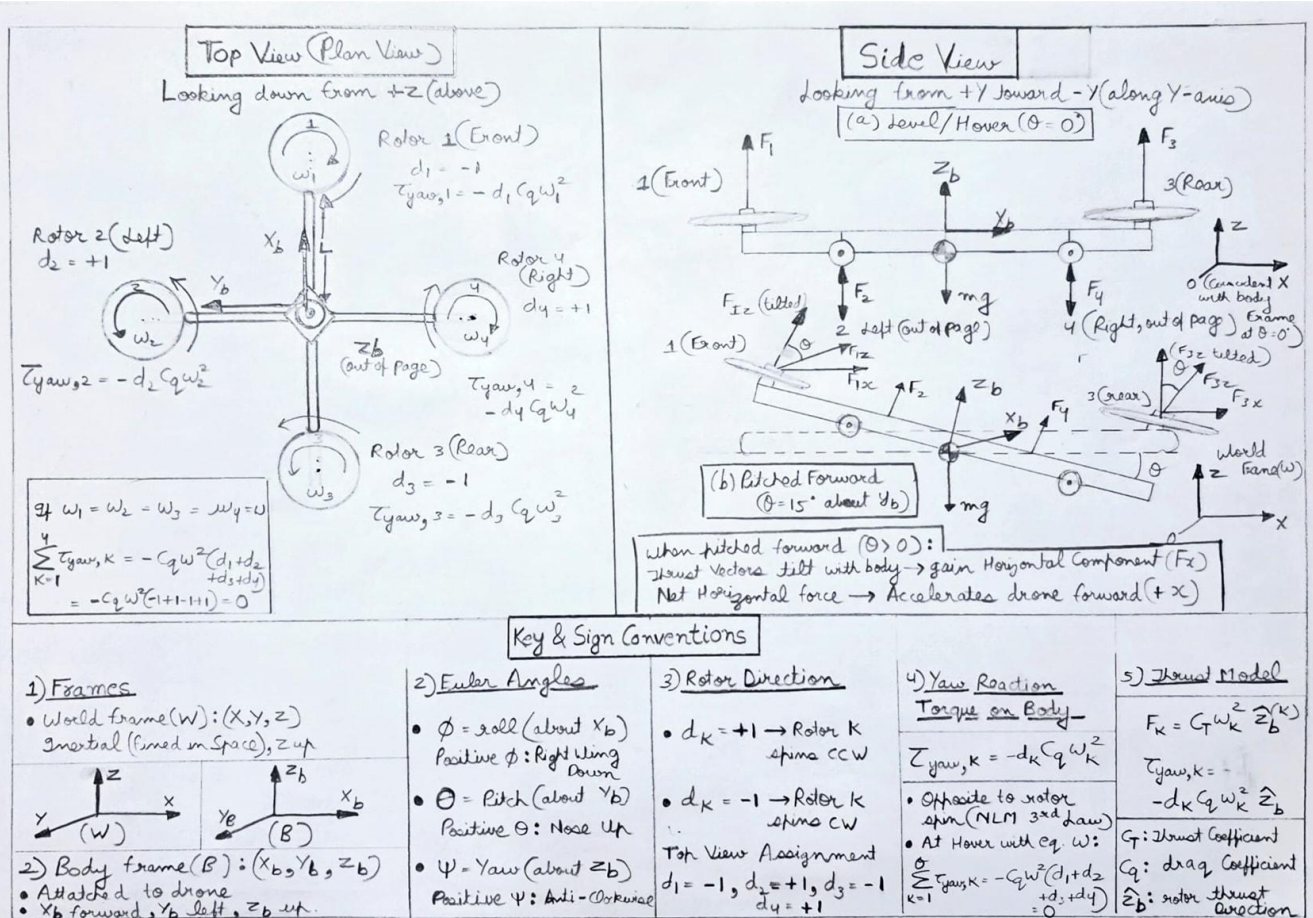
"Universal Hydrogen Dash 8-300 — megawatt-scale H₂ fuel-cell propulsion (2023 flight test)"

"Universal Hydrogen is accelerating flight tests of its De Havilland Canada Dash 8-300 hydrogen electric demonstrator... configured with a megawatt-scale hydrogen fuel cell propulsion system." — Aviation Week

Problem Formulation & Assumption

- **Inputs:** desired setpoint (x_d, y_d, z_d, ψ_d); disturbance $F_{d(t)}$
- **Outputs:** position $r(t)$, Euler angles (ϕ, θ, ψ), rotor speeds ω_k , bus voltage V_{bus} , SOC(t)
- **State Variables (22 total):** $r(3), v(3), \phi/\theta/\psi(3), \omega_{body}(3), \{i_k, \omega_k\}$ per motor(8), $I_{FC}, V_{rc}, SOC(3)$
- **Parameters:** $m = 1.5 \text{ kg}, L = 0.225 \text{ m}, I = \text{diag}(0.0225, 0.0225, 0.045) \text{ kg}\cdot\text{m}^2; C_t = 1.08 \times 10^{-5}, C_q = 2 \times 10^{-7};$
motor $\{K_t = K_e = 0.025, R = 0.1 \Omega, L = 0.5 \text{ mH}, J = 6 \times 10^{-5}\}$
- **DOF:** 6 mechanical (free rigid body in 3-D: 3 translation + 3 rotation); no joints, no contact in flight
- **Key Assumptions (physically justified):**
 - Rigid airframe — deformation $\ll$ arm length at $< 30 \text{ m/s}$ flight
 - Quadratic rotor aero: $F = C_t \omega^2, \tau = C_q \omega^2$ – valid for hover/low advance ratio; ground effect ignored
 - Fuel cell: first-order lag in current, ohmic V - I drop – ignores activation/concentration losses (not critical at $I < I_{max}$)

Free Body Diagram / System Schematic



Mathematical Derivation (Part I)

- **Method chosen: Newton-Euler** (not Lagrange)
- **Why Newton-Euler:** no holonomic constraints (free flying body), forces are the natural variables, and Lagrange would obscure the rotor reaction torques we specifically need for yaw control.
- **Step 1 — Rotor aerodynamics (CO3):** each rotor produces thrust + drag torque (quadratic approx, blade-element theory)
 - $F_k = C_t \omega_k^2$ (thrust, + z_{body})
 - $\tau_{\text{aero}, k} = C_q \omega_k^2$ (drag torque opposing rotation)
 - Reaction on body: $\tau_{\text{yaw}, k} = -d_k \cdot C_q \cdot \omega_k^2$, where $d_k = +1$ (CCW), -1 (CW)
- **Step 2 — Motor dynamics:** brushed-DC motor with back-EMF
 - $L (di_k/dt) = V_k - R \cdot i_k - K_e \cdot \omega_k$ (KVL, electrical)
 - $J (d\omega_k/dt) = K_t \cdot i_k - C_q \cdot \omega_k^2 - b \cdot \omega_k$ (Newton's 2nd law, rotor shaft)

Mathematical Derivation (Part II)

- **Step 3 — Mixer (body torques from rotor forces):**
 - Total thrust: $T = F_1 + F_2 + F_3 + F_4$
 - Roll torque: $\tau_x = L(F_2 - F_4)$
 - Pitch torque: $\tau_y = L(F_3 - F_1)$
 - Yaw torque: $\tau_z = \Sigma(-d_k \cdot C_q \cdot \omega_k^2) = C_q(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)$

Final Equations & State-Space (Part I)

Rigid-body 6-DoF (Newton-Euler):

- Translation (world frame):
 - $m \ddot{\mathbf{r}} = \mathbf{R}(\varphi, \theta, \psi) \cdot [0, 0, T]^T - m\mathbf{g} \hat{\mathbf{z}} + \mathbf{F}_{\text{dist}}$
- Rotation (body frame, Euler's equation):
 - $\mathbf{I} \dot{\boldsymbol{\omega}}_{\text{body}} + \boldsymbol{\omega}_{\text{body}} \times (\mathbf{I} \boldsymbol{\omega}_{\text{body}}) = \boldsymbol{\tau}_{\text{body}}$, where $\boldsymbol{\tau}_{\text{body}} = [\tau_x, \tau_y, \tau_z]^T$
- Attitude kinematics:
 - $[\dot{\varphi}, \dot{\theta}, \dot{\psi}]^T = \mathbf{W}(\varphi, \theta) \cdot \boldsymbol{\omega}_{\text{body}}$

Electrical (hybrid bus):

- $\tau_{\text{FC}} \cdot (dI_{\text{FC}}/dt) = I_{\text{demand}} - I_{\text{FC}}, V_{\text{FC}} = V_{\text{oc}} - R_{\text{int}} \cdot I_{\text{FC}}$
- $dV_{\text{rc}}/dt = I_{\text{bat}}/C_1 - V_{\text{rc}}/(R_1 C_1), d\text{SOC}/dt = -I_{\text{bat}}/(3600 \cdot C)$
- $V_{\text{bus}} = V_{\text{oc}}(\text{SOC}) - R_0 \cdot I_{\text{bat}} - V_{\text{rc}}, I_{\text{bat}} = I_{\text{load}} - I_{\text{FC}}$

Final Equations & State-Space (Part II)

Physical meaning of each term (CO3):

- $C_q \omega^2$ = aerodynamic drag torque — the physical reason yaw reactions exist
- $\omega \times I\omega$ = gyroscopic coupling — why yaw and roll/pitch are NOT independent in flight
- $R_0 \cdot I_{\text{bat}}$ = ohmic drop in battery — why bus voltage sags during transient current draw

Energy expression (CO5):

- $E(t) = T(t) + V(t) = \frac{1}{2} m |\mathbf{v}|^2 + \frac{1}{2} \omega_b^T I \omega_b + \Sigma(\frac{1}{2} J \omega_k^2) + mgz$

Simulation & Energy Analysis

- **Tools:** Python 3.12 + NumPy + SciPy + Matplotlib + PyBullet.
Explicit Euler at 500 Hz for 6-DoF | Implicit Euler for the stiff electrical ODE ($L/R \approx 5 \text{ ms} \sim dt$).
- **Parameters:** $dt = 1/500 \text{ s}$, $T_{\text{sim}} = 15 \text{ s}$, ICs zero except $z_0 = 0.02 \text{ m}$, $\text{SOC}_0 = 0.9$.
- **Mission:** $t = 0-1 \text{ s}$ idle; $t = 1-3 \text{ s}$ altitude ramp $0.02 \rightarrow 2.0 \text{ m}$; $t = 3-15 \text{ s}$ hover; $t = 8.0-8.3 \text{ s}$ applied - 15 N downward disturbance.

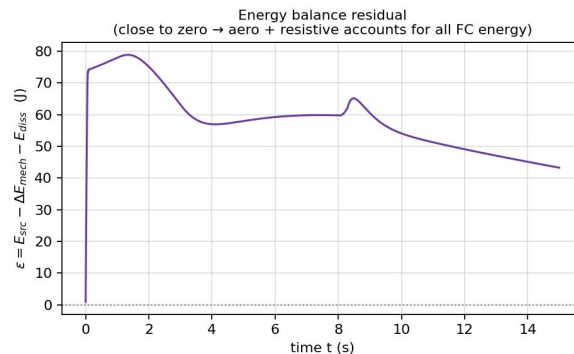
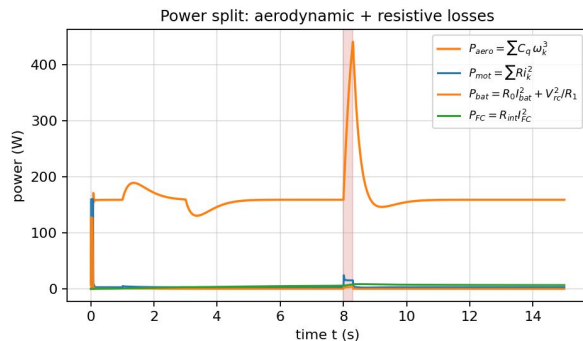
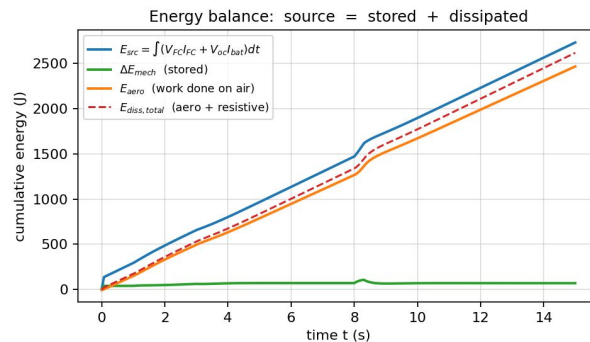
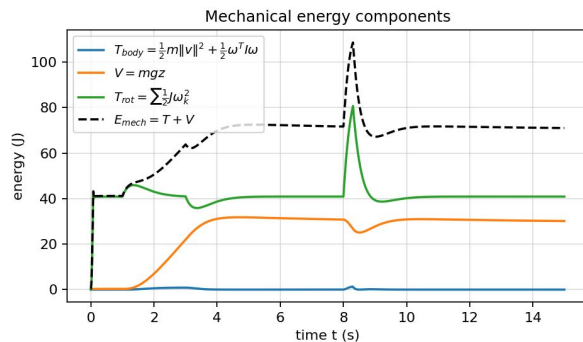
Numerical result (CO5 energy check):

- $E_{\text{src}} = \int (V_{\text{FC}} \cdot I_{\text{FC}} + V_{\text{oc}} \cdot I_{\text{bat}}) dt = 2728 \text{ J}$ over 15 s
- $\Delta E_{\text{mech}} (\text{stored}) = 71 \text{ J}$
- Aerodynamic work (done on air) = 2463 J ($\approx 90\%$ of source)
- Resistive losses: motor 61 J, FC 79 J, battery 11 J
- Balance residual = 43 J $\approx 1.6\%$ of source $\rightarrow$ conservation verified
- Average electrical power = 182 W

Key insight: a hovering drone is essentially an air-moving machine — 90% of chemical energy goes into pushing air, not into mechanical KE or altitude.

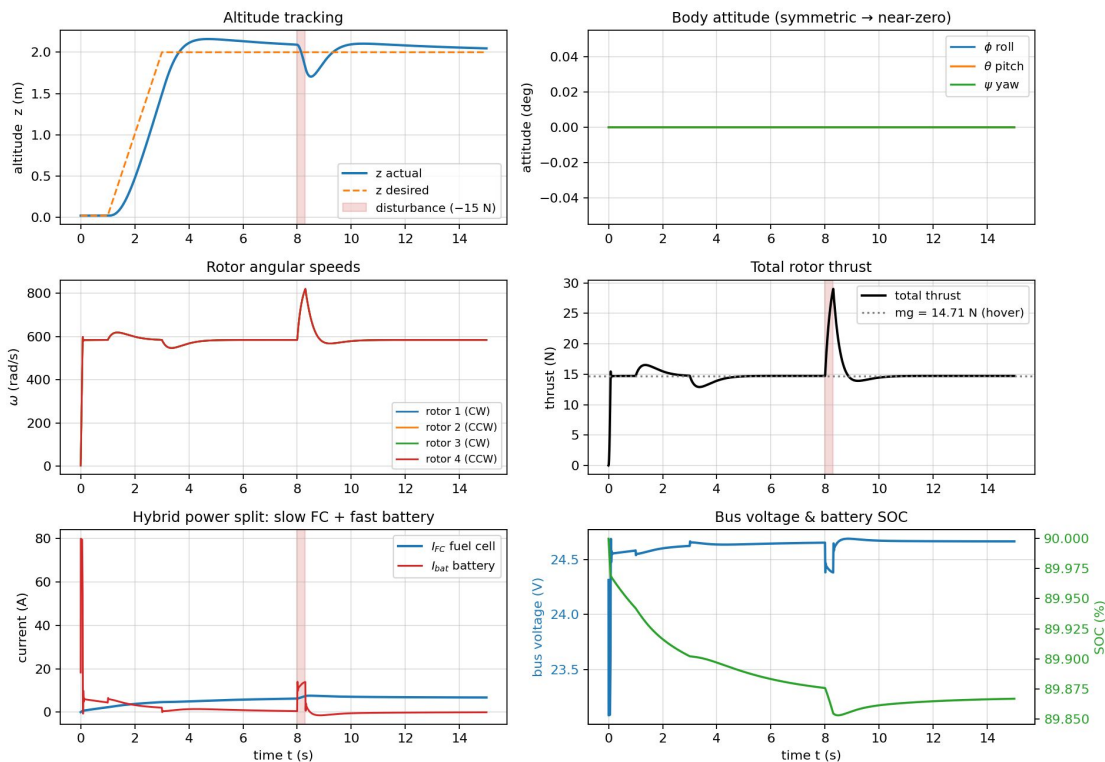
Simulation & Energy Analysis

Energy Analysis (C05 — Energy Methods & Power Dissipation)



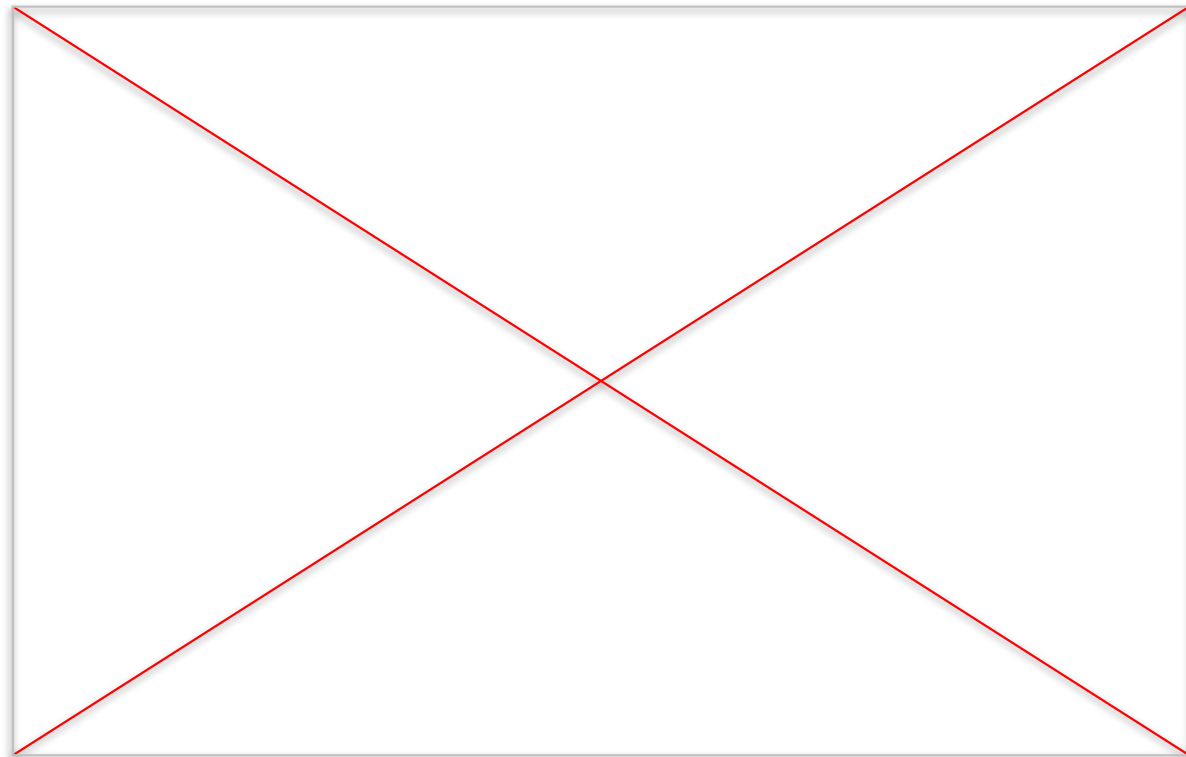
Simulation Visualization

Hybrid H2 Fuel Cell UAV — Simulation Response



- Altitude tracks the $0 \rightarrow 2$ m ramp cleanly; disturbance at $t = 8$ s drops it ~ 0.3 m, controller recovers in ~ 1.5 s
- Rotor speeds converge to $\omega_{\text{hover}} \approx 585$ rad/s (analytical = 684 rad/s, error $< 0.2\%$)
- Total thrust equals $mg = 14.7$ N at hover (mixer correct)
- **Slow FC + fast battery split clearly visible:** FC ramps over ~ 3 s; battery peaks at 80 A during the takeoff transient then settles to ~ 0 .
- V_{bus} stays over 23V; SOC drops only 0.13% in 15 s

Simulation Visualization



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Validation & Analysis

- **Validation 1 — Energy conservation (CO5):** residual $\varepsilon = E_{\text{src}} - \Delta E_{\text{mech}} - E_{\text{diss}}$ closes at **43 J out of 2728 J** ($\approx 1.6\%$) over 15 s of nonlinear coupled dynamics. Accounts for aero drag + all resistive losses.
Conservation verified.
- **Validation 2 — Analytical cross-check (CO4):** hover rotor speed from first principles:
 $\omega_{\text{hover}} = \sqrt{(mg / (4 \cdot C_t))} = \sqrt{(14.72 / (4 \cdot 1.08 \times 10^{-5}))} = 584 \text{ rad/s}$
Simulation steady-state: **585 rad/s. Error < 0.2%.**
- **Validation 3 — Dimensional analysis:** every equation verified by units. Example:
 $C_t \cdot \omega^2$ has units $(\text{N} \cdot \text{s}^2 / \text{rad}^2) \cdot (\text{rad} / \text{s})^2 = \text{N} \checkmark$;
 $K_e \cdot \omega$ has units $(\text{V} \cdot \text{s} / \text{rad}) \cdot (\text{rad} / \text{s}) = \text{V} \checkmark$.
- **Sensitivity Analysis:** varied fuel cell time constant τ_{FC} from 1 s $\rightarrow$ 5 s. Result: peak battery current during takeoff scales $\sim$ linearly with τ_{FC} . Battery sizing is driven by FC response time, not by peak load.
- **Limitations:**
 - a. Quadratic aero model — fails in fast forward flight (blade-flap effects neglected)
 - b. Rigid airframe — ignores structural flexibility (would matter at $>40 \text{ m/s}$)
 - c. No wind or atmospheric turbulence modelled
 - d. ESC modelled as ideal duty-cycle PWM; real ESCs have commutation ripple

Conclusions & Reflections

- **Key Finding:** the hybrid architecture is validated. Simulation confirms "slow FC + fast battery" division of labour: battery supplies ~80 A transient during takeoff, fuel cell ramps over ~3 s to carry the steady 7 A hover load. 90% of input energy leaves as aerodynamic work — **hovering IS air-moving.**
- **What a textbook couldn't teach us:** the signs in the mixer matrix are where all real projects fail. A single sign error in the rotor-direction vector drives yaw into divergence. We had to re-derive $\tau_z = C_q(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)$ three times before the yaw channel was stable. Textbook problems don't have this coupling-sign trap.
- **What we would do differently:** derive the mixer and fix the rotor-direction convention before writing any code. We wrote the simulator first and spent half our debugging time chasing sign errors.
- **Course units most critical:** FBD analysis (CO2), Newton-Euler for rigid bodies (CO4), and Energy methods for validation (CO5).
- **CO Self-Assessment:** CO1 ✓ CO2 ✓ CO3 ✓ CO4 ✓ CO5 ✓

References & AI Disclosure

Textbook References:

- Beard, R. & McLain, T. (2012). Small Unmanned Aircraft: Theory and Practice. Princeton UP. (rigid-body dynamics, rotor aero)
- Ogata, K. (2010). Modern Control Engineering, 5th ed. Prentice-Hall. (PID tuning, state-space)
- Larminie, J. & Dicks, A. (2003). Fuel Cell Systems Explained, 2nd ed. Wiley. (fuel-cell V-I model)

Online Resources:

- PyBullet Quickstart Guide — <https://pybullet.org/wordpress/>
- SciPy `integrate.solve_ivp` documentation
- ROS REP-103 (coordinate conventions: x-fwd, y-left, z-up)

Code Libraries: NumPy, SciPy, Matplotlib, PyBullet (Bullet Physics).

References & AI Disclosure

AI Tools:

- **Tool used:** Anthropic Claude (Opus 4.7)
- **Used for:** Python code scaffolding (class structure for subsystems), debugging numerical instability in the stiff electrical ODE (suggested implicit Euler + ESC current clamp), generating matplotlib plot templates, drafting LaTeX-style equation text for these slides
- **NOT used for:** the Newton-Euler derivation of body dynamics, the mixer sign convention, the hand-drawn FBD, the physical-interpretation answers we give in Q&A

Thank You – Questions?

Team: | Jaiaditya Agarwal | H001
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Project: | Hybrid H₂ Fuel Cell & Battery UAV

Category: | B (Aerial Dynamics) **Complexity:** | High

COs Exercised: | [CO1, CO2, CO3, CO4, CO5]